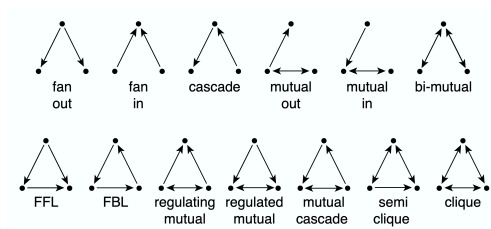


L5: Directed Subgraph Connectivity Patterns



So far, we have primarily focused on clustering in undirected networks. Such **undirected clustering** manifests through the presence of triangles of connected nodes. What about directed networks, however?

In that case, we can have various **types** of connected triplets of nodes. The upper part of the visualization shows the 13 different types of connection patterns between three weakly connected network nodes.

Note that each of the 13 patterns is distinct when we consider the directionality of the edges. Also, if you are not convinced that these are the only 13 possible patterns you can try to find any additional patterns (*hint: you will not be able to!*).

Each of these patterns is also given a name (e.g., **FeedForward Loop or FFL**). Instead of using the word "pattern", we will be referring to such small subgraph types

as **network motifs**.

A specific network motif (e.g., **FFL**) may occur in a network just based on chance. How can we check if that motif occurs much more frequently than that? What does it mean when a specific network motif occurs very frequently? Or the opposite, what does it mean when a network motif occurs much less frequently than expected based on chance? We will answer these questions next.

Food For Thought

Given that we know that there are 13 motifs between 3 weakly connected nodes, how many network motifs exist between 4 weakly connected nodes?

